

COMP-2540: MIDTERM EXAM 1 (15%)



The duration of the exam is 70 Minutes. This is a close-book exam. You are not allowed to use notes, calculators, computers, or cell phones. Please select the most appropriate answer for each question. Remember to fill in the scantron sheet using #2 pencil.

Questions 1-5 are related to our Assignment One. Suppose that there are n tokens and the vocabulary size is m .

Q1. What is the task of assignment 1?

- (A) Compare sorting algorithms
- (B) Reverse strings
- (C) Three-way disjointness
- (D) Count word frequency
- (E) Element uniqueness problem

Q2. What is the time complexity for the program `count_LINKED_LIST_BAD`?

- (A) $O(n)$
- (B) $O(n \log n)$
- (C) $O(nm)$
- (D) $O(nmm)$
- (E) $O(nnm)$

Q3. What should be the time complexity for `count_LINKED_LIST_GOOD`?

- (A) $O(n)$
- (B) $O(n \log n)$
- (C) $O(nm)$
- (D) $O(nmm)$
- (E) $O(nnm)$

Q4. What is the time complexity for `count_ARRAY`?

- (A) $O(n)$
- (B) $O(n \log n)$
- (C) $O(nm)$
- (D) $O(nmm)$
- (E) $O(nnm)$

Q5. What is the time complexity if you sort the array first, then count the frequency?

- (A) $O(n)$
- (B) $O(n \log n)$
- (C) $O(nm)$
- (D) $O(nmm)$
- (E) $O(nnm)$

Q6. Given the following recursive program,

```
boolean x(int[] data, int target, int low, int high) {
    if (low > high) return false;
    int mid = (low + high) / 2;
    if (target == data[mid]) return true;
    if (target < data[mid])
        return x(data, target, low, mid - 1);
    return x(data, target, mid + 1, high);
}
```

It's recurrence relation is

- (A) $T(n) = T(n - 1) + 1$
- (B) $T(n) = T(n - 1) + n$
- (C) $T(n) = T(n/2) + 1$
- (D) $T(n) = 2T(n/2) + n$
- (E) $T(n) = T(n - 1) + T(n - 2) + 1$

Q7. For the above program, it's time complexity is

- (A) $O(n)$
- (B) $O(n^2)$
- (C) $O(\log n)$
- (D) $O(n \log n)$
- (E) $O(2^n)$

Q8. Consider an array of size 10 (indices are from 0 to 9). The recursive algorithm for reversing an array studied in class, namely `reverseArray(A, i, j)`, makes the following recursive calls (in that order):

```
void reverseArray(int[] data, int low, int high) {
    if (low < high) {
        int temp = data[low];
        data[low] = data[high];
        data[high] = temp;
        reverseArray(data, low + 1, high - 1);
    }
}
```

- (a) `reverseArray(A, 4, 6)`, `reverseArray(A, 3, 7)`, `reverseArray(A, 2, 8)`, etc.
- (b) `reverseArray(A, 0, 9)`, `reverseArray(A, 1, 8)`, `reverseArray(A, 2, 7)`, etc.
- (c) `reverseArray(A, 0, 1)`, `reverseArray(A, 0, 2)`, `reverseArray(A, 0, 3)`, etc.
- (d) `reverseArray(A, 9, 8)`, `reverseArray(A, 7, 6)`, `reverseArray(A, 5, 4)`, etc.

Q9. The following sequence is equal to

$$1 + 2 + 3 + \dots + n =$$

- (a) $2n$ (d) $n^2/2 + n/2$
 (b) n^2
 (c) $n^2/2$ (e) $n(n-1)/2$

Q10. Which is the closest to the following sequence

$$\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$$

- (a) $2n$ (d) $\ln n$
 (b) $n \log n$
 (c) $\log n$ (e) n

Q11. The following sequence is equal to

$$1 + 2^1 + 2^2 + \cdots + 2^{n-1}$$

- (a) 2^n (d) $n^2 + n$
 (b) n^2
 (c) $n^2 - 1$ (e) $2^n - 1$

Q12. Given the following algorithm to find the max element from an array.

```
int arrayMax(int[] data) {
    int n=data.length;
    int currentMax=data[0];
    for (int j=1;j<n;j++)
        if(data[j]>currentMax)
            currentMax=data[j];
    return currentMax;
}
```

How many times the assignment statement

$currentMax = data[j]$ is executed on average?

- (a) n (d) $n \log n$
 (b) $n/2$
 (c) $\ln n$ (e) $\log n$

Q13. The following program calculate the **prefix average** as we discussed in class. Namely, given a sequence x consisting of n numbers, we want to compute a sequence a such that a_j is the average of elements $x_0, \dots, x_j$, for $j = 0, \dots, n-1$, that is,

$$a_j = \frac{\sum_{i=0}^j x_i}{j+1} \quad (1)$$

```
double[] prefixAverage(double[] x) {
    int n = x.length;
    double[] a = new double[n];
    for (int j=0; j < n; j++) {
        double total = 0;
        for (int i=0; i <= j; i++)
            total += x[i];
        a[j] = total / (j+1);
    }
    return a;
}
```

The code can be improved so that the better code has time complexity

- (a) $O(n)$ (d) $O(n \log n)$
 (b) $O(n^2)$
 (c) $O(\log n)$ (e) $O(2^n)$

Q14. Which of the following is a disadvantage of using arrays over linked lists?

- (A) Arrays require contiguous memory allocation.
 (B) The size of an array is fixed after declaration.
 (C) Insertion and deletion at arbitrary positions are slow.
 (D) All of the above.

Q15. Which of the following is NOT true about a singly linked list?

- (A) The elements are stored in non-contiguous memory locations.
 (B) Each node has a pointer to the next node in the sequence.
 (C) Insertion at the head takes $O(1)$ time.
 (D) Accessing the n -th element takes $O(1)$ time.

Q16. What is the time complexity of searching for an element in an unsorted array of size n ?

- (A) $O(1)$ (D) $O(n \log n)$
 (B) $O(\log n)$
 (C) $O(n)$ (E) $O(n^2)$

Q17. What is the time complexity of searching for an element in a sorted array of size n using binary search?

- (A) $O(1)$ (D) $n \log n$
 (B) $O(\log n)$
 (C) $O(n)$ (E) $O(n^2)$

Q18. What is the worst-case time complexity of deleting the tail node in a singly linked list with n nodes?

- (A) $O(1)$ (D) $O(n \log n)$
 (B) $O(\log n)$
 (C) $O(n)$ (E) $O(n^2)$

Q19. What is the time complexity of the following program:

```
public static boolean x(int[] data) {
    int n = data.length;
    int[] temp = Arrays.copyOf(data, n);
    Arrays.sort(temp);
    for (int j=0; j<n-1; j++)
        if (temp[j] == temp[j+1])
            return false;
    return true;
}
```

- (A) $O(\log n)$ (D) $O(n^2)$
 (B) $O(n)$ (E) $O(n^3)$
 (C) $O(n \log n)$

- (A) 1 (D) 4
 (B) 2
 (C) 3 (E) 5

Q20. What is the time complexity for the following code:

```
boolean x(int[] groupA, int[] groupB, int[] groupC) {
    for (int a : groupA)
        for (int b : groupB)
            if (a == b)
                for (int c : groupC)
                    if (a == c)
                        return false;
    return true;
}
```

- (A) $O(\log n)$ (D) $O(n^2)$
 (B) $O(n)$
 (C) $O(n \log n)$ (E) $O(n^3)$

Q21. What is the time complexity for the following program:

```
public static String aFunction(char c, int n) {
    String answer = "";
    for (int j=0; j < n; j++)
        answer += c;
    return answer;
}
```

- (A) $O(1)$ (D) $O(n \log n)$
 (B) $O(\log n)$
 (C) $O(n)$ (E) $O(n^2)$

Q22. Given the following merge sort code, which line has an error?

```
1 <K>void mergeSort(K[] S, Comparator<K> comp){
2     int n = S.length;
3     if (n < 2) return;
4     int mid = n/2;
5     K[] S1 = Arrays.copyOfRange(S, 0, mid);
6     K[] S2 = Arrays.copyOfRange(S, mid+1, n);
7     mergeSort(S1, comp);
8     mergeSort(S2, comp);
9     merge(S1, S2, S, comp);
10 }
```

- (A) 3 (D) 6
 (B) 4
 (C) 5 (E) 7

Q23. Which line of the following code has an error?

```
1 int x(int n) {
2     if (n==0)
3         return 1;
4     else
5         return n * x(n);
6 }
```

Q24. What is the time complexity for the following code?

```
int x(int[] data, int low, int high) {
    if (low > high) return 0;
    if (low == high) return data[low];
    int mid = (low + high) / 2;
    return x(data, low, mid) + x(data, mid+1, high);
}
```

- (A) $O(1)$ (D) $n \log n$
 (B) $O(\log n)$
 (C) $O(n)$ (E) $O(n^2)$

Q25. Which of the following operations on an array has a time complexity of $O(1)$ in the worst case?

- (A) Searching for an element.
 (B) Deleting an element.
 (C) Inserting an element at a given index.
 (D) Accessing an element at a specific index.

Q26. Which of the following is true for doubly linked lists but not for singly linked lists?

- (A) They allow traversal in both forward and backward directions.
 (B) Each node has a pointer to the next node.
 (C) Insertion at the head is $O(1)$.
 (D) Deletion of the head node takes $O(1)$ time.

Q27. What is the time complexity of inserting an element into an unsorted array of size n ?

- (A) $O(1)$ (D) $n \log n$
 (B) $O(\log n)$
 (C) $O(n)$ (E) $O(n^2)$

Q28. Given functions $f(n) = 3n^2 - 2n + 2 \log n$ and $g(n) = 2n^2 + 10n + 2n \log n$, which of the following is correct?

- (A) f grows asymptotically faster than g
 (B) f grows asymptotically slower than g
 (C) Both $f(n)$ and $g(n)$ are $O(n^2)$
 (D) $g(n)$ is $\Theta(n^2)$ but $f(n)$ is not

Q29. In a singly linked list discussed in class:

- (A) Insertion at the tail must be done in $O(n)$
 (B) Removing at the tail can be done in $O(1)$
 (C) Removing at the head can be done in $O(1)$
 (D) Finding the node in the middle of the list takes $O(\log n)$

Q30. Given the recurrence relation $T(n) = 2T(n/2) + n$. What is the closed-form upper bound solution for $T(n)$?

- (A) $O(n)$ (D) $O(\log n)$
 (B) $O(n^2)$
 (C) $O(n \log n)$ (E) $O(2^n)$

Q31. Given the recurrence relation $T(n) = T(n/2) + 1$. What is the closed-form upper bound solution for $T(n)$?

- (A) $O(n)$ (D) $O(\log n)$
 (B) $O(n^2)$
 (C) $O(n \log n)$ (E) $O(2^n)$

Q32. Given the recurrence relation $T(n) = 2T(n-1) + 1$. What is the closed-form upper bound solution for $T(n)$?

- (A) $O(n)$ (D) $O(\log n)$
 (B) $O(n^2)$
 (C) $O(n \log n)$ (E) $O(2^n)$

Q33. Given the recurrence relation $T(n) = T(n-1) + 1$. What is the closed-form upper bound solution for $T(n)$?

- (A) $O(n)$ (D) $O(\log n)$
 (B) $O(n^2)$
 (C) $O(n \log n)$ (E) $O(2^n)$

Q34. Given the following program, what is the recurrence relation for the running time of the program?

```
long x(int n) {
    if (n <= 1) return n;
    return x(n-2) + x(n-1);
}
```

- (A) $T(n) = T(n-1) + 1$
 (B) $T(n) = T(n-1) + n$
 (C) $T(n) = T(n/2) + 1$
 (D) $T(n) = 2T(n/2) + n$
 (E) $T(n) = T(n-1) + T(n-2) + 1$

Q35. For the above program, which of the following best describes its time complexity?

- (A) $O(n)$ (D) $O(\log n)$
 (B) $O(n^2)$
 (C) $O(n \log n)$ (E) $O(2^n)$

Q36. Given the following program, what is the recurrence relation for the running time of the program?

```
int x(int[] data, int low, int high) {
    if (low > high) return 0;
    if (low == high) return data[low];
    int mid = (low + high) / 2;
    return x(data, low, mid) + x(data, mid+1, high);
}
```

- (A) $T(n) = T(n-1) + 1$
 (B) $T(n) = T(n-1) + n$
 (C) $T(n) = 2T(n/2) + 1$
 (D) $T(n) = 2T(n/2) + n$
 (E) $T(n) = T(n-1) + T(n-2) + 1$

Q37. What is the time complexity of the Selection Sort algorithm, supposing that the data size is n ?

- (A) $O(n)$ (D) $O(n \log n)$
 (B) $O(n^2)$
 (C) $O(\log n)$ (E) $O(2^n)$

Q38. What is the time complexity of the Insertion Sort algorithm, supposing that the data size is n ?

- (A) $O(n)$ (D) $O(n \log n)$
 (B) $O(n^2)$
 (C) $O(\log n)$ (E) $O(2^n)$

Q39. What is the time complexity of the following recursive function?

```
int someFunction(int n) {
    if (n == 0) return 1;
    else return n * someFunction(n-1);
}
```

- (A) $O(n)$ (D) $O(\log n)$
 (B) $O(2^n)$
 (C) $O(n^2)$ (E) $O(n \log n)$

Q40. What is the time complexity of the following function?

```
public static long[] fibonacci(int n) {
    if (n <= 1) {
        long[] answer = {n, 0};
        return answer;
    } else {
        long[] temp = fibonacci(n-1);
        long[] answer = {temp[0] + temp[1], temp[0]};
        return answer;
    }
}
```

- (A) $O(1)$ (D) $O(n^2)$
 (B) $O(n \log n)$
 (C) $O(n)$ (E) $O(2^n)$

Q41. What is the time complexity of the following recursive function?

```
int x(int n):
    if (n <= 1) return 1;
    return 3 * x(n - 1)
```

- (A) $O(n \log n)$ (D) $O(n^3)$
 (B) $O(3^n)$
 (C) $O(n)$ (E) $O(\log n)$

Q42. Which of the following algorithm discussed in class uses dynamic programming algorithm design strategy?

- (A) Merge sort (D) Binary search
 (B) Insertion sort
 (C) Fibonacci algorithm (E) Selection sort

Q43. Which of the following algorithm discussed in class uses divide and conquer algorithm design strategy?

- (A) Merge sort (D) Binary search
 (B) Insertion sort
 (C) Fibonacci algorithm (E) Selection sort

Q44. Given the following algorithm, suppose that $S1=[1,3,6]$, and $S2=[2,3,4,5]$.

```
public static void f(int[] S1, int[] S2, int[] S) {
    int i = 0, j = 0;
    while (i + j < S.length) {
        if (j == S2.length
            || (i < S1.length && S1[i] < S2[j]))
            S[i+j] = S1[i++];
        else S[i+j] = S2[j++];
    }
}
```

What is the value in array S after the execution of the algorithm?

- (A) 1 3 6 2 3 4 5 (C) 2 3 4 5
 (B) 1 3 6 (D) 1 2 3 3 4 5 6

Q45. In our context, the base of $\log n$ is

- (a) 2 (d) 4
 (b) 10
 (c) e (e) 1